
Observation-Aware Video World Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

Modern image-to-video generators are increasingly used as video world models: given an anchor image and a task description, they infer a latent trajectory for the future scene. We argue that a central limitation of this setting is that the rollout is usually open-loop. Once generation begins, the model must maintain geometry, identity, motion, and physical plausibility using only its learned prior, without an observation-feedback mechanism that can correct the evolving latent state. Under this view, failures of motion smoothness, object persistence, and physical behavior are not only visual artifacts; they are state-estimation errors in the latent world-model rollout. This paper proposes retrieval-guided data assimilation as a control-theoretic paradigm for observation-aware video world models. For each image-prompt query, a LanguageBind retrieval memory selects one semantically related real-video trajectory from an OpenVid-style bank. The retrieved trajectory is encoded as an observation sequence and supplied to a fine-tuned Video-as-Prompt in-context DiT pathway. Generated latent tokens are treated as the forecast state, retrieved video tokens as external observation evidence, and the forecast-side attention output as the analysis state used for noise prediction. A controlled Moving MNIST latent-ETKF study and a six-case Wan2.1-14B qualitative pilot provide initial evidence for the claim: retrieved observations can increase scene-level motion in favorable cases, while revealing the remaining challenges of blur, framing drift, retrieval mismatch, and runtime.

1 Introduction

A useful video world model must do more than produce visually plausible frames. Following the “Trinity of Consistency” view of general world models, it should satisfy three coupled requirements: modal consistency, which aligns language, images, and video with the same semantic state; spatial consistency, which preserves the geometry, layout, and identity of the world; and temporal consistency, which makes the state evolve causally over time [23]. This trinity is a sharper starting point for our problem than generic video quality. If a generated video violates spatial consistency, objects can drift, pop, or reappear with different geometry. If it violates temporal consistency, motion becomes jittery or physically implausible. If it violates modal consistency, the video may be smooth but no longer follow the prompt or anchor image.

Recent video world-model work has made the spatial part of this trinity especially visible. World-consistent diffusion models introduce explicit 3D coordinate supervision to make generated frames geometrically coherent [27]. GEN3C uses a 3D cache to avoid object popping and improve world-consistent camera-controlled generation [15]. Long-term spatial memory methods explicitly store and retrieve 3D scene memory because limited temporal context causes forgetting during revisits [25], while compressed memory approaches such as WorldPack improve spatial consistency by packing and retrieving past trajectories [13]. These works point to a common diagnosis: current video generators are not yet reliable world models because their latent state is weakly anchored to persistent spatial evidence and real dynamical observations.

However, modern text/image-to-video models typically roll out the world state without real-time observations. The model receives an anchor image and a text prompt, then predicts a video consistent with that initial state and intent [20]. This interface gives a strong modal condition, but once generation begins, spatial and temporal consistency are left to the model’s internal prior. The model must simulate dynamics from limited conditioning, without access to external observations of how similar scenes usually evolve. The result can be temporally irregular motion, weak object persistence, and physically implausible state transitions. We therefore take the following position: image-to-video generation should be treated as open-loop latent world-model forecasting, and a core missing primitive is observation feedback for correcting the forecast state.

Inspired by classical data assimilation (DA), where a model forecast is corrected by external observations to produce an analysis state, we transfer the same logic into the DiT denoiser of an image-to-video world model. The noisy generated latent tokens are the forecast state of the world model, the single retrieved real-video token sequence is an external observation of plausible spatial-temporal evolution, and cross attention between them is the learned analysis operator that updates the representation used for noise prediction. In this framing, spatial consistency, motion smoothness, and physics fidelity become state-estimation properties: generated tokens should remain compatible with real observed dynamics in the retrieval memory. The main claim of this paper is that retrieval should not only be used as semantic context; it can serve as an empirical observation source for correcting latent world-model rollouts.

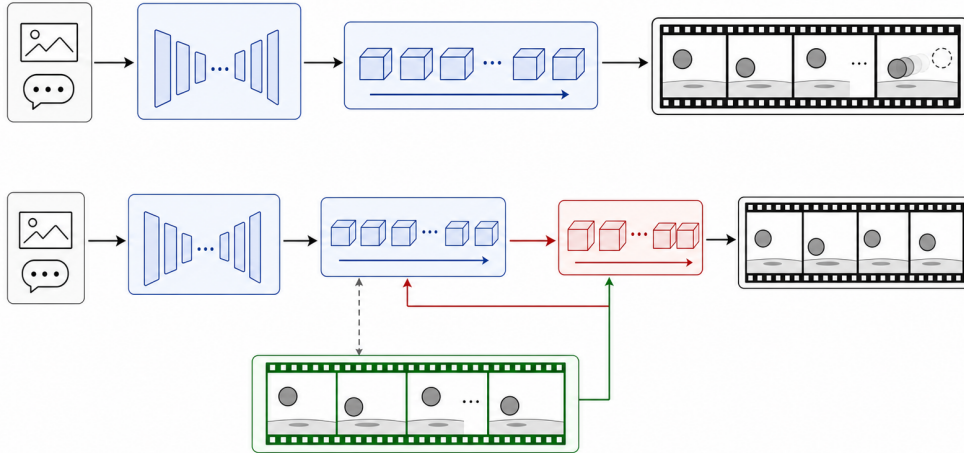


Figure 1: Conceptual contrast between open-loop video generation and retrieval-guided observation correction. The top row illustrates the conventional image-to-video setting: the model performs an open-loop world-model forecast from the prompt and anchor image, so spatial drift and physically weak motion must be handled by the model prior alone. The bottom row illustrates our control-theoretic paradigm: a semantically related real-video trajectory is retrieved as external observation evidence, then used to form an observation-feedback analysis update that corrects the forecast state before video prediction.

Our current contributions are:

1. We formulate prompt-conditioned image-to-video generation as open-loop latent world-model forecasting and identify the absence of observation feedback as a structural source of drift and weak dynamics.
2. We introduce retrieval-guided data assimilation: a retrieved real-video trajectory is treated as an external observation of plausible spatial-temporal evolution, not merely as semantic conditioning.
3. We instantiate this idea through a fine-tuned Video-as-Prompt in-context DiT pathway, interpreting forecast-to-observation attention as an analysis update without adding a new attention mechanism.

4. We provide preliminary controlled and qualitative evidence that observation feedback can improve scene-level motion while exposing the spatial fidelity and runtime limitations that remain.

2 Preliminaries and Preliminary Study

2.1 Data-Assimilation Intuition

Data assimilation can be understood as a feedback correction rule. A model first predicts a *forecast* state, an observation arrives, and the difference between what is observed and what was expected becomes the *innovation*. In the scalar Kalman setting,

$$x^a = x^f + \frac{P^f}{P^f + R} (y - x^f), \quad (1)$$

where x^f is the forecast, y is the observation, P^f is forecast uncertainty, R is observation-noise variance, and x^a is the corrected analysis state. The gain term finds the proper weight between prediction and observation so that the next predicted state has minimum expected error relative to the unknown ground truth. ETKF extends this idea from one state to an ensemble of high-dimensional states, updating the ensemble mean and spread in a low-dimensional ensemble space rather than forming a full covariance matrix. Appendix A gives the full ETKF notation and the one-dimensional intuition figure.

2.2 Preliminary Moving MNIST Study

We start with a simple Moving MNIST video predictor to depict the idea of data assimilation in a controlled video setting [19]. The host model rolls out future frames from a latent recurrent state, while the DA variant applies a latent ETKF update whenever a noisy full-frame observation is revealed. We use this study to measure the information gap between the model-only forecast and the observation-corrected analysis. Let $p_f(z)$ be the forecast distribution and $p_a(z) = p(z | Y)$ be the analysis distribution after observation Y . Relative entropy $D_{\text{KL}}(p_a \| p_f)$ measures the information discrepancy between analysis and forecast, while residual mutual information measures how much observation information remains outside the corrected state. In the ideal ground-truth limit, both the divergence from reality and the residual observation information would approach zero.

The Moving MNIST results support the state-estimation interpretation used in the rest of the paper. Increasing the ETKF ensemble from 8 to 500 members reduces relative entropy from 295.4926 to 19.6006 and residual mutual information from 0.5709 to 0.0622, showing that the analysis state moves closer to the observed rollout as the assimilation update becomes stronger. The cost is computational, because stronger explicit ETKF correction requires larger ensembles. This motivates using DiT in-context attention as a single-path approximation to the forecast–observation–analysis update in the retrieval-guided video model. Detailed equations, setup, and the full table and plot are provided in Appendix A.

3 Method

3.1 Overview

To generalise the data-assimilation method to a video world model, we use a fine-tuned Video-as-Prompt-style pathway [1] as the host image-text-to-video system and interpret its in-context pathway as an attention-based approximation to a DA analysis update. The base image-to-video branch is the forecast stream. The retrieved reference-video branch is the observation stream. The in-context DiT pathway allows these streams to interact without explicitly sampling a large ETKF ensemble. The pipeline is organized into four steps:

1. retrieve one real trajectory from the shared LanguageBind world memory;
2. encode the selected trajectory as observation tokens O ;
3. run fine-tuned Video-as-Prompt inference with noisy forecast tokens H^f and observation tokens H^o ;

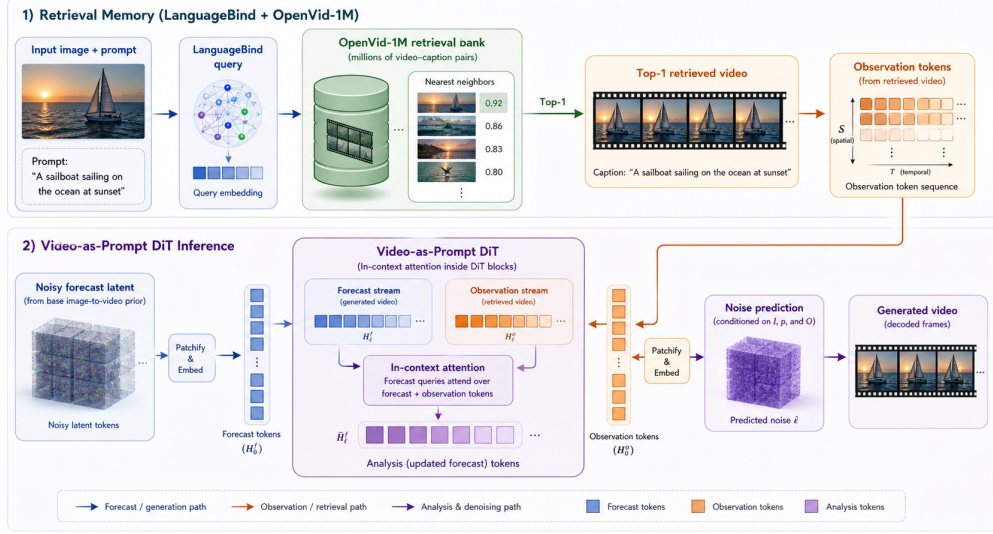


Figure 2: Architecture of the retrieval-guided fine-tuned Video-as-Prompt inference pipeline. The input image and text prompt form a LanguageBind query over an OpenVid-1M retrieval bank. The most relevant retrieved clip is encoded as observation tokens, while the base image-to-video branch supplies noisy forecast tokens. Inside the Video-as-Prompt DiT denoiser, forecast tokens attend to retrieved observation tokens; the forecast-side analysis tokens are then used for noise prediction and video generation.

4. interpret the forecast-side in-context output as analysis tokens before noise prediction.

Figure 2 summarizes the retrieval and DiT inference architecture.

3.2 Retrieval Memory and Shared Embedding Space

We formulate the retrieval bank as

$$\mathcal{R} = \{(r_i, c_i)\}_{i=1}^N, \quad (2)$$

where r_i is a real video trajectory and c_i is its caption. A LanguageBind encoder [30] maps text, image, and video inputs into a shared embedding space. Given a prompt p and an anchor image I , we build the retrieval query

$$q = Q_{\text{LB}}(p, I), \quad (3)$$

and retrieve the single nearest clip

$$(r^*, c^*) = \arg \max_{(r_i, c_i) \in \mathcal{R}} s(q, r_i). \quad (4)$$

This retrieval memory defines the external observation source. The retrieved clip is not a label and not a target future. It is an empirical trajectory from world memory: one example of how a semantically related scene evolves in real video.

3.3 Video World Model Setting

Let an image-to-video world-model input be an anchor observation and a task descriptor

$$I \in \mathcal{I}, \quad p \in \mathcal{P}. \quad (5)$$

The model samples a latent trajectory and decodes it as a video

$$\hat{x} = (\hat{x}_1, \dots, \hat{x}_T) \quad (6)$$

from a noisy latent state z_t^p , the prompt p , and the anchor image I . Here, z_t^p is the native DiT latent state at denoising time t , and the superscript p denotes the prompt/image-conditioned prior rollout before any retrieved-video observation is used. We define the corresponding DA forecast state as

$$z_t^f \equiv z_t^p. \quad (7)$$

The superscript f denotes the forecast state: before assimilation, the world-model forecast is exactly the base DiT latent generated from (I, p) . The retrieved real-video clip is an observation of plausible motion and physical evolution for a semantically related scene. The goal is not to reconstruct a specific target clip, but to produce an analysis trajectory z_t^a whose decoded video is more stable, smoother, and more physically faithful than the unassimilated forecast.

In world-model terms, retrieval-guided inference is a latent state-estimation problem with an external memory. The base DiT defines a transition forecast from (I, p) to a noisy latent trajectory. Retrieval provides a single empirical observation trajectory O showing how a similar state evolves. The desired update is an analysis operator

$$z_t^a = \mathcal{A}_\theta(z_t^f, O, p, I), \quad (8)$$

where z_t^a is the analysis latent state after incorporating the retrieved observation at denoising step t ; $\mathcal{A}_\theta$ is the analysis operator induced by the fine-tuned Video-as-Prompt in-context DiT pathway with parameters θ ; z_t^f is the forecast latent state from the base image-to-video DiT before observation use; O is the single retrieved real-video observation-token sequence; p is the text prompt; and I is the input anchor image. The analysis state should preserve the semantic intent of the prompt while improving temporal coherence and physical plausibility.

3.4 Observation Tokens

The retrieved video trajectory is converted into one token sequence rather than a single summary vector or a multi-sample set. Let E_{vid} denote the frozen video preprocessing, VAE encoding, and patch embedding path that maps the retrieved clip into DiT-compatible spatio-temporal tokens. For the most similar retrieved clip, we write the observation token as:

$$O = E_{\text{vid}}(r^*, c^*), \quad (9)$$

where the caption c^* supplies text-side observation context. We do not concatenate multiple retrieved clips or use retrieval weights as attention biases in this first version. The key point is that the retrieved video enters the DiT as observation data for the world state, rather than as an external score or auxiliary label.

3.5 Forecast and Observation Streams

At denoising step t , the base DiT receives a noisy latent video z_t^f together with prompt and image conditioning. After patch embedding, this produces forecast tokens

$$H_0^f = P_{\text{patch}}(z_t^f, I, p). \quad (10)$$

The retrieved video branch receives the observation tokens O and optional observation text/image embeddings, giving

$$H_0^o = P_{\text{obs}}(O). \quad (11)$$

The forecast stream is the internal world-model rollout. The observation stream is a real-video branch. Both streams are propagated through the DiT, but only the forecast-side analysis output is used to predict the noise for the generated video.

3.6 Data-Assimilation View of In-Context Attention

For a selected DiT block ℓ , let H_ℓ^f denote forecast tokens and H_ℓ^o denote observation tokens. In the Video-as-Prompt-style in-context pathway, the generated-video branch and reference-video branch can be written with separate QKV projections:

$$Q_\ell^f, K_\ell^f, V_\ell^f = W_{qkv}^f H_\ell^f, \quad Q_\ell^o, K_\ell^o, V_\ell^o = W_{qkv}^o H_\ell^o. \quad (12)$$

The in-context full-attention form is

$$\begin{bmatrix} \bar{H}_\ell^f \\ \bar{H}_\ell^o \end{bmatrix} = \text{Attn} \left(\begin{bmatrix} Q_\ell^f \\ Q_\ell^o \end{bmatrix}, \begin{bmatrix} K_\ell^f \\ K_\ell^o \end{bmatrix}, \begin{bmatrix} V_\ell^f \\ V_\ell^o \end{bmatrix} \right), \quad (13)$$

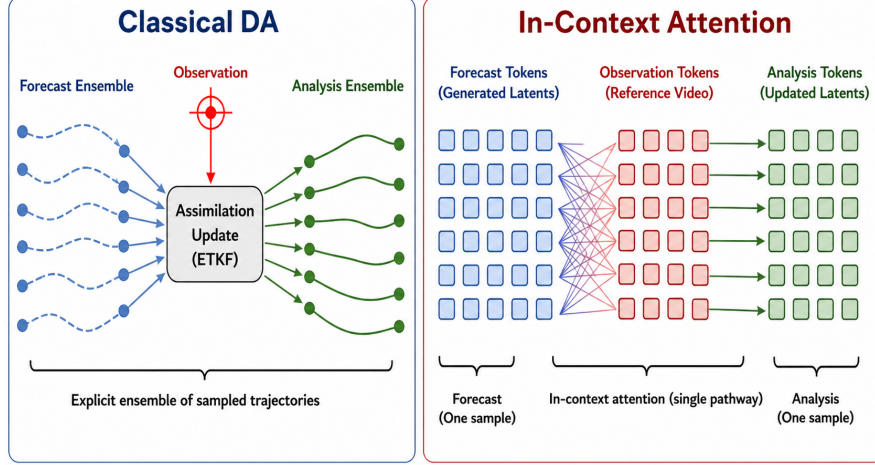


Figure 3: Conceptual comparison between explicit classical DA and in-context attention. Classical DA updates an ensemble of forecast trajectories with an observation, while in-context attention lets generated forecast tokens attend to retrieved observation tokens inside one DiT pathway.

where $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d})V$. For a lighter cross-attention implementation, the forecast stream can query both streams while the observation stream is left fixed:

$$\bar{H}_\ell^f = \text{Attn}\left(Q_\ell^f, \begin{bmatrix} K_\ell^f \\ K_\ell^o \end{bmatrix}, \begin{bmatrix} V_\ell^f \\ V_\ell^o \end{bmatrix}\right). \quad (14)$$

In both cases, the forecast-side output $\bar{H}_\ell^f$ is the analysis representation. The cross terms between Q_ℓ^f and K_ℓ^o measure which observation tokens are relevant to the current generated latent forecast. We interpret these weights as an in-context score for motion smoothness and physical compatibility.

3.7 Noise Prediction with Observation Tokens

After the in-context interaction, the DiT predicts the noise using the updated forecast stream:

$$\hat{\epsilon} = \epsilon_\theta(z_t^f, t, p, I, O). \quad (15)$$

The observation tokens are therefore part of the DiT world-model input. They do not produce a separate output video; they influence the generated trajectory by modifying the forecast stream during attention.

3.8 Fine-Tuning Objective and Data

Following the Video-as-Prompt paradigm [1], we fine-tune only the retrieval-guided in-context pathway while keeping the base image-to-video DiT, VAE, and text encoder frozen. For each captioned target clip $x = (x_1, \dots, x_T)$, we use the first frame $I = x_1$ and caption p , and retrieve a non-identical clip r^* with caption c^* as the reference observation, excluding source duplicates and near duplicates. With $z_0 = E_{\text{vid}}(x)$ and noised latent z_t , the objective is the standard conditional video diffusion loss $\mathcal{L}_{\text{ft}} = \mathbb{E}_{x, I, p, r^*, c^*, t, \epsilon} [\|\epsilon - \epsilon_\theta(z_t, t, I, p, r^*, c^*)\|_2^2]$, where r^* is both the Video-as-Prompt reference video and, in our DA interpretation, the observation trajectory used to correct forecast tokens during denoising.

4 Experiment

Retrieval bank and algorithm. OpenVid-1M is used as the external observation memory [12]. Each candidate clip is represented by cached LanguageBind caption and video embeddings; for each test input, the prompt and anchor image are embedded in the same space and the nearest real clip is selected as the observation trajectory. Only the most relevant retrieved clip is used, with no averaging, weighting, or post-hoc reranking.

Inference with retrieved video. We compare Wan2.1-14B image-to-video inference with the retrieval-guided path. The baseline uses only the input image and prompt, while our path also supplies the retrieved clip and input image as Video-as-Prompt references in the fine-tuned pathway [1]. Both paths generate 17 frames with identical settings, testing whether real-video observations improve motion smoothness, physical consistency, and scene-level dynamics.

4.1 Qualitative Experiment

To study the effect of retrieved observation on video generation, we compare the baseline and retrieval-guided path on six manually supplied image–video examples. Each case shows the retrieved reference, baseline output, and retrieved-video output under the same prompt and seed. Figure 4 shows representative cases in the main text; the remaining temporal panels and the compact six-case table are moved to Appendix B to keep the main submission within the page limit.

Qualitative temporal samples: cases 1-2

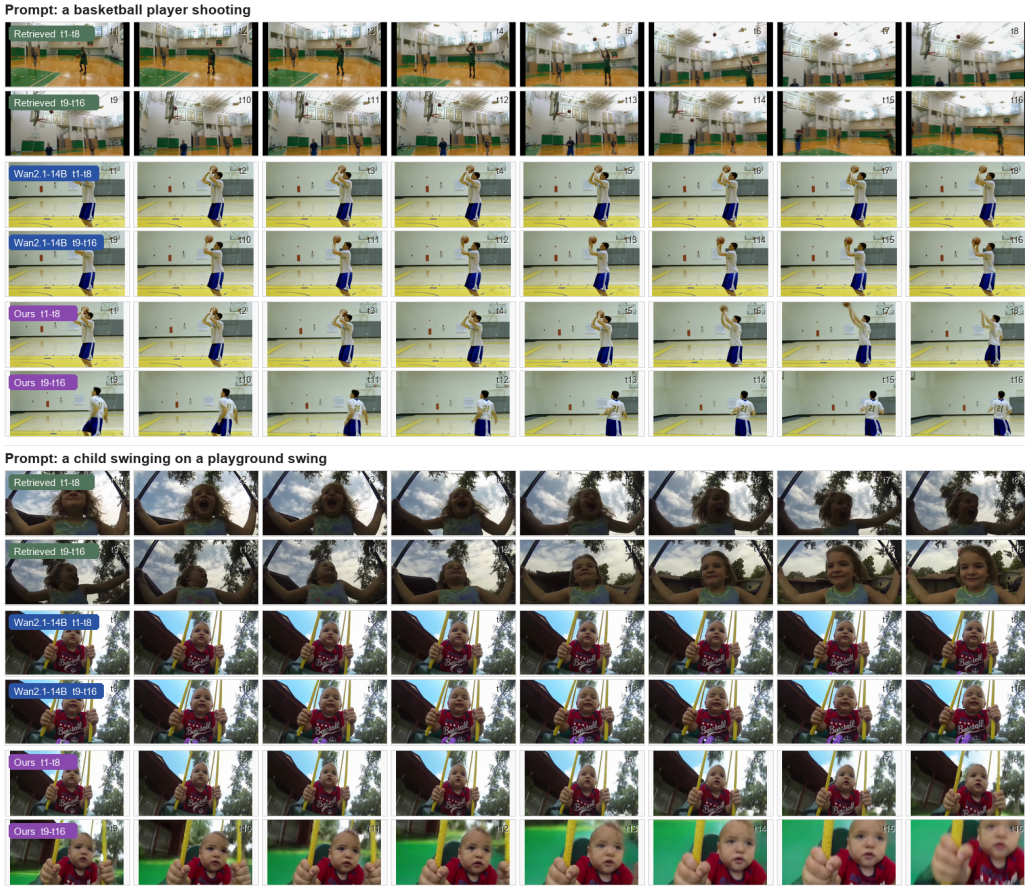


Figure 4: Temporal-sample qualitative comparison for cases 1–2, showing 16 normalized samples from the retrieved reference, baseline, and retrieved-video outputs.

In cases 1–2, the retrieved observation increases visible motion but also exposes the main tradeoff. The basketball example gains a clearer shooting trajectory, while the swing example gains stronger depth and camera motion at the cost of blur and weaker identity preservation.

Analysis. Across the six cases, the retrieved observation changes not only the amount of motion, but also the consistency between the input prompt and the generated dynamics. The baseline often preserves the first-frame layout and prompt semantics, yet under-realizes the mechanics: basketball shooting has limited body displacement, dog fetching remains nearly stationary, and running leaves much of the road and crowd static. These videos are spatially stable but physically conservative,

because short prompts do not specify contact timing, depth change, camera parallax, or body-state transitions. The retrieved clip supplies this missing spatial-temporal evidence without requiring a detailed prompt. In favorable cases, the output shows clearer follow-through, swing depth, approach motion, surf and road parallax, and global scene motion, making the generated video closer to real footage than the baseline. The remaining challenge is spatial alignment with the anchor image: camera viewpoint, subject scale, identity, and background correspondence can still introduce blur, framing drift, or appearance change when the retrieved clip is imperfectly matched. These issues are likely reducible through stronger data processing, such as viewpoint-aware retrieval, identity and background filtering, and camera-motion normalization. We leave this direction as future work, while the present result shows that observation already improves physical consistency under prompt under-specification.

5 Discussion

The qualitative study and the Moving MNIST analysis point to the same interpretation: observation-aware video generation is useful when the generator must infer realistic dynamics from limited prompt evidence.

Advantages. The main advantage of the proposed formulation is that it gives image-to-video generation a clear state-estimation structure. The base video generator supplies a forecast prior, the retrieval bank supplies an observation trajectory, and the fine-tuned Video-as-Prompt pathway supplies the analysis update. This gives the video world model a feedback channel that is absent from ordinary open-loop generation. A model conditioned only on an initial image and prompt may be visually strong, but once rollout begins it has no explicit mechanism for comparing its predicted trajectory with external evidence. In our formulation, retrieval memory acts as a nonparametric world memory, and the analysis update is the mechanism by which the latent forecast is corrected.

This advantage is especially visible for short or underspecified prompts. Prompts such as “a dog fetching a ball” or “a basketball player shooting” specify the semantic event, but leave the dynamics under-specified: they do not determine contact timing, object acceleration, body pose transitions, camera motion, or interaction geometry. In this motion-underconstrained setting, an image-to-video model may preserve the first-frame appearance while producing conservative motion or physically weak state transitions. The retrieved video reduces this under-specification by supplying an empirical trajectory of how a similar event unfolds. Thus the relevant gain is not only stronger motion, but improved physical plausibility under prompt under-specification. Figure 3 also explains why the implementation uses in-context attention rather than explicit ETKF at video scale: the Moving MNIST study shows that larger ETKF ensembles reduce the forecast–analysis information gap, but the runtime ratio increases rapidly, while the DiT attention pathway approximates the forecast–observation–analysis update in one denoising route.

Disadvantages and future improvements. The main disadvantage is that observation-aware motion is not automatically observation-faithful motion. If the retrieved trajectory has strong motion but imperfect spatial correspondence to the input image, the generated video may gain dynamics while losing sharpness, identity preservation, or framing stability. This failure mode separates temporal improvement from full world-model consistency: a useful analysis update must preserve modal consistency with the prompt, spatial consistency with the anchor image, and temporal consistency with the retrieved dynamics. Future work can address this limitation through stronger data processing, including viewpoint-aware retrieval, identity and background filtering, camera-motion normalization, and retrieval confidence thresholds before the observation is supplied to the generator.

Additional future work is required. The observation-conditioned Video-as-Prompt path is slower than optimized single-pass generation; future systems should tune the forecast-to-observation attention strength and reduce runtime overhead. Because attention is not a calibrated uncertainty estimate, evaluation should formulate information-theoretic metrics for video world models, measuring entropy, relative entropy, and mutual information between the forecast, retrieved observation, and analysis trajectory. These metrics would quantify the information gap closed by observation-aware correction. Finally, the spatial state should be made explicitly three-dimensional through geometry-aware retrieval, camera pose, depth, or 3D scene memory, reducing the bias between 2D appearance tokens and the 3D mechanics required for physical and spatial consistency.

6 Conclusion

This paper argues that image-to-video generation is best viewed not only as conditional rendering, but as latent world-model forecasting. Under that view, a central weakness of current systems is their open-loop nature: after receiving the image and prompt, they must maintain spatial and temporal consistency without external correction. Retrieval-guided data assimilation addresses this weakness by treating generated latent tokens as a forecast state and retrieved real-video tokens as an observation trajectory. Their interaction inside the fine-tuned Video-as-Prompt DiT pathway is interpreted as an analysis update used for noise prediction. The resulting claim is simple: retrieval should be an observation-feedback mechanism for correcting world-model rollouts, not only a source of semantic conditioning.

Future work. The Moving MNIST study provides controlled evidence that a DA-style update can improve dynamical rollout quality when observations are available. The six-case qualitative study shows the corresponding behavior in a large image-to-video system: retrieved observations can increase dynamic motion, but they can also introduce blur, layout drift, and additional runtime. Future work should run a controlled same-backbone comparison, replace manually supplied reference clips with automatic LanguageBind retrieval from the cached OpenVid memory, evaluate more prompts and random seeds, and add human or metric-based judgments of motion smoothness and physical consistency. A stronger implementation should also expose or tune the forecast-to-observation attention strength, reduce runtime, and preserve the motion benefit while avoiding the spatial drift observed in the pilot.

References

- [1] Yuxuan Bian, Xin Chen, Zenan Li, Tiancheng Zhi, Shen Sang, Linjie Luo, and Qiang Xu. Video-as-prompt: Unified semantic control for video generation. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=8FihPljvWf>. arXiv:2510.20888.
- [2] Craig H. Bishop, Brian J. Etherton, and Sharanya J. Majumdar. Adaptive sampling with the ensemble transform kalman filter. part i: Theoretical aspects. *Monthly Weather Review*, 129(3):420–436, 2001. doi: 10.1175/1520-0493(2001)129<0420:ASWTET>2.0.CO;2.
- [3] Caterina Buizza, César Quilodrán Casas, Philip Nadler, Julian Mack, Stefano Marrone, Zainab Titus, Clémence Le Cornec, Evelyn Heylen, Tolga Dur, Luis Baca Ruiz, et al. Data learning: Integrating data assimilation and machine learning. *Journal of Computational Science*, 58: 101525, 2022. doi: 10.1016/j.jocs.2021.101525.
- [4] Siyi Chen, Yixuan Jia, Qing Qu, He Sun, and Jeffrey A. Fessler. Flowdas: A flow-based framework for data assimilation. *arXiv preprint arXiv:2501.16642*, 2025. doi: 10.48550/arXiv.2501.16642. URL <https://arxiv.org/abs/2501.16642>.
- [5] Taiye Chen, Xun Hu, Zihan Ding, and Chi Jin. Learning world models for interactive video generation. *arXiv preprint arXiv:2505.21996*, 2025. doi: 10.48550/arXiv.2505.21996. OpenReview id FzfYoUp8F1.
- [6] Yixiang Chen, Peiyan Li, Jiabing Yang, Keji He, Xiangnan Wu, Yuan Xu, Kai Wang, Jing Liu, Nianfeng Liu, Yan Huang, and Liang Wang. Bridgev2w: Bridging video generation models to embodied world models via embodiment masks. *arXiv preprint arXiv:2602.03793*, 2026. doi: 10.48550/arXiv.2602.03793.
- [7] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. doi: 10.48550/arXiv.1803.10122. URL <https://arxiv.org/abs/1803.10122>.
- [8] Muiyang He, Hanzhong Guo, Junxiong Lin, and Yizhou Yu. Video generation models as world models: Efficient paradigms, architectures and algorithms. *arXiv preprint arXiv:2603.28489*, 2026. doi: 10.48550/arXiv.2603.28489.
- [9] Daniel Hodyss and Matthias Morzfeld. Using diffusion models to do data assimilation. *arXiv preprint arXiv:2506.02249*, 2025. doi: 10.48550/arXiv.2506.02249.
- [10] Yinan Huang, Hans Hao-Hsun Hsu, Junran Wang, Bo Dai, and Pan Li. Accelerated sequential flow matching: A bayesian filtering perspective. *arXiv preprint arXiv:2602.05319*, 2026. doi: 10.48550/arXiv.2602.05319.
- [11] Peter Manshausen, Yair Cohen, Peter Harrington, Jaideep Pathak, Mike Pritchard, Piyush Garg, Morteza Mardani, Karthik Kashinath, Simon Byrne, and Noah Brenowitz. Generative data assimilation of sparse weather station observations at kilometer scales. *arXiv preprint arXiv:2406.16947*, 2024. doi: 10.48550/arXiv.2406.16947.
- [12] Kepan Nan, Rui Xie, Penghao Zhou, Tiehan Fan, Zhenheng Yang, Zhijie Chen, Xiang Li, Jian Yang, and Ying Tai. Openvid-1m: A large-scale high-quality dataset for text-to-video generation. In *The Thirteenth International Conference on Learning Representations*, 2025. doi: 10.48550/arXiv.2407.02371. URL <https://openreview.net/forum?id=j7kdXSrISM>. OpenReview id j7kdXSrISM.
- [13] Yuta Oshima, Masahiro Suzuki, Yusuke Iwasawa, Yutaka Matsuo, and Hiroki Furuta. World-pack: Compressed memory improves spatial consistency in video world modeling. *arXiv preprint arXiv:2512.02473*, 2025. doi: 10.48550/arXiv.2512.02473. OpenReview id CuNH3zxgm.
- [14] Elia Peruzzo, Dejia Xu, Xingqian Xu, Humphrey Shi, and Nicu Sebe. Ragme: Retrieval augmented video generation for enhanced motion realism. *arXiv preprint arXiv:2504.06672*, 2025. doi: 10.48550/arXiv.2504.06672. URL <https://arxiv.org/abs/2504.06672>.

- [15] Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, and Jun Gao. Gen3c: 3d-informed world-consistent video generation with precise camera control. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6121–6132, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Ren_GEN3C_3D-Informed_World-Consistent_Video_Generation_with_Precise_Camera_Control_CVPR_2025_paper.html.
- [16] Jonas Scholz and Richard E. Turner. Warm starts accelerate conditional diffusion. *arXiv preprint arXiv:2507.09212*, 2025. doi: 10.48550/arXiv.2507.09212.
- [17] Yu Shang, Lei Jin, Yiding Ma, Xin Zhang, Chen Gao, Wei Wu, and Yong Li. Longscape: Advancing long-horizon embodied world models with context-aware moe. *arXiv preprint arXiv:2509.21790*, 2025. doi: 10.48550/arXiv.2509.21790.
- [18] Xingjian Shi, Zhoung Chen, Hao Wang, Dit-Yan Yeung, Wai-kin Wong, and Wang-chun Woo. Convolutional lstm network: A machine learning approach for precipitation nowcasting. In *Advances in Neural Information Processing Systems*, volume 28, pages 802–810, 2015.
- [19] Nitish Srivastava, Elman Mansimov, and Ruslan Salakhudinov. Unsupervised learning of video representations using lstms. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 843–852. PMLR, 2015.
- [20] Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. doi: 10.48550/arXiv.2503.20314. URL <https://arxiv.org/abs/2503.20314>.
- [21] Jiahao Wang, Luoxin Ye, Taiming Lu, Junfei Xiao, Jiahao Zhang, Yuxiang Guo, Xijun Liu, Rama Chellappa, Cheng Peng, Alan Yuille, and Jieneng Chen. Evoworld: Evolving panoramic world generation with explicit 3d memory. *arXiv preprint arXiv:2510.01183*, 2025. doi: 10.48550/arXiv.2510.01183.
- [22] Luozhou Wang, Zhifei Chen, Yihua Du, Dongyu Yan, Wenhao Ge, Guibao Shen, Xinli Xu, Leyi Wu, Man Chen, Tianshuo Xu, Peiran Ren, Xin Tao, Pengfei Wan, and Ying-Cong Chen. A mechanistic view on video generation as world models: State and dynamics. *arXiv preprint arXiv:2601.17067*, 2026. doi: 10.48550/arXiv.2601.17067.
- [23] Jingxuan Wei, Siyuan Li, Yuhang Xu, Zheng Sun, Junjie Jiang, Hexuan Jin, Caijun Jia, Honghao He, Xinglong Xu, Xi Bai, et al. The trinity of consistency as a defining principle for general world models. *arXiv preprint arXiv:2602.23152*, 2026. doi: 10.48550/arXiv.2602.23152.
- [24] Fan Wu, Jiacheng Wei, Ruibo Li, Yi Xu, Junyou Li, Deheng Ye, and Guosheng Lin. Ic-world: In-context generation for shared world modeling. *arXiv preprint arXiv:2512.02793*, 2025. doi: 10.48550/arXiv.2512.02793.
- [25] Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025. doi: 10.48550/arXiv.2506.05284. URL <https://arxiv.org/abs/2506.05284>.
- [26] Jiahao Zhang, Muqing Jiang, Nanru Dai, Taiming Lu, Arda Uzunoglu, Shunchi Zhang, Yana Wei, Jiahao Wang, Vishal M. Patel, Paul Pu Liang, Daniel Khashabi, Cheng Peng, Rama Chellappa, Tianmin Shu, Yilun Du, and Jieneng Chen. World-in-world: World models in a closed-loop world. *arXiv preprint arXiv:2510.18135*, 2025. doi: 10.48550/arXiv.2510.18135.
- [27] Qihang Zhang, Shuangfei Zhai, Miguel Angel Bautista, Kevin Miao, Alexander Toshev, Joshua Susskind, and Jiatao Gu. World-consistent video diffusion with explicit 3d modeling. *arXiv preprint arXiv:2412.01821*, 2024. doi: 10.48550/arXiv.2412.01821. URL <https://arxiv.org/abs/2412.01821>.
- [28] Hongkai Zheng, Wenda Chu, Austin Wang, Nikola Kovachki, Ricardo Baptista, and Yisong Yue. Ensemble kalman diffusion guidance: A derivative-free method for inverse problems. *Transactions on Machine Learning Research*, 2025. doi: 10.48550/arXiv.2409.20175. URL <https://arxiv.org/abs/2409.20175>. arXiv:2409.20175.

- [29] Siyuan Zhou, Yilun Du, Yuncong Yang, Lei Han, Peihao Chen, Dit-Yan Yeung, and Chuang Gan. Learning 3d persistent embodied world models. In *Advances in Neural Information Processing Systems*, 2025. OpenReview id XTTbzC7O2T.
- [30] Bin Zhu, Bin Lin, Munan Ning, Yang Yan, Jiaxi Cui, HongFa Wang, Yatian Pang, Wenhao Jiang, Junwu Zhang, Zongwei Li, Wancai Zhang, Zhifeng Li, Wei Liu, and Li Yuan. Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment. In *International Conference on Learning Representations*, 2024. doi: 10.48550/arXiv.2310.01852. URL <https://openreview.net/forum?id=QmZKc7UZCy>.

A Detailed Preliminaries and Moving MNIST Study

A.1 Kalman and ETKF Details

For a scalar state, let x^f be the model forecast, y be the observed position, and let the observation directly measure the same state. The scalar Kalman update is

$$d = y - x^f, \quad (16)$$

$$K = \frac{P^f}{P^f + R}, \quad (17)$$

$$x^a = x^f + Kd, \quad (18)$$

where d is the innovation, P^f is the forecast uncertainty, R is the observation-noise variance, K is the Kalman gain, and x^a is the corrected analysis state. The purpose of the Kalman gain is to find the proper weight between prediction and observation so that the corrected state has minimum expected error relative to the unknown ground-truth state.

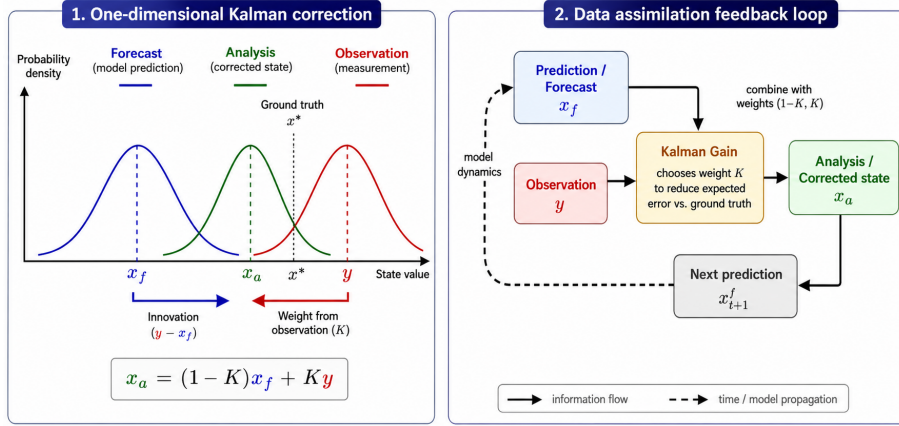


Figure 5: One-dimensional Kalman filtering intuition for data assimilation. The forecast and observation are weighted by the Kalman gain to form an analysis state closer to the unknown ground truth x_* in expected squared error.

ETKF extends the same idea to an M -member high-dimensional ensemble [2]. Let $\{x_i^f\}_{i=1}^M$ be forecast states, with ensemble mean $\bar{x}^f$ and anomaly matrix $X'^f = [x_1^f - \bar{x}^f, \dots, x_M^f - \bar{x}^f]$. Given an observation operator H , observation covariance R , and observation y , the observation-space anomalies are $Y'^f = [H(x_1^f) - \bar{y}^f, \dots, H(x_M^f) - \bar{y}^f]$. ETKF computes

$$S = (M - 1)I + (Y'^f)^\top R^{-1} Y'^f, \quad (19)$$

$$w^a = S^{-1} (Y'^f)^\top R^{-1} (y - \bar{y}^f), \quad (20)$$

$$T^a = [(M - 1)S^{-1}]^{1/2}, \quad (21)$$

and updates the ensemble by $\bar{x}^a = \bar{x}^f + X'^f w^a$ and $X'^a = X'^f T^a$. Here S measures the forecast uncertainty in observation space, w^a gives the correction weights, and T^a updates the ensemble spread.

A.2 Moving MNIST Information Metrics

The Moving MNIST experiment uses the standard 10-to-10 prediction setup [19]. The first ten frames form the context, and OpenLoopVideoPredictor rolls out the next ten frames through a residual convolutional encoder-decoder and ConvLSTM-style transition module [18]. The DA variant keeps the same checkpoint and changes only the test-time state update: it perturbs the latent recurrent state (z_t, h_t, c_t) to form an ensemble, propagates the ensemble through the same forecast step, and applies the latent ETKF update whenever a noisy full-frame observation is revealed.

Let Z denote the latent world-model state and Y denote the observation. The forecast distribution $p_f(z)$ is the model’s belief before seeing the observation, while the analysis distribution $p_a(z) = p(z | Y)$ is the observation-corrected belief. Relative entropy measures the information discrepancy between the analysis and forecast,

$$D_{\text{KL}}(p_a \| p_f) = \int p_a(z) \log \frac{p_a(z)}{p_f(z)} dz, \quad (22)$$

and mutual information measures how much information the observation carries about the latent world state,

$$I(Z; Y) = H(Z) - H(Z | Y). \quad (23)$$

Table 1: Preliminary Moving MNIST latent-DA information metrics. Lower relative entropy and lower residual mutual information indicate a smaller information gap after assimilation.

Method	Relative entropy ↓	Mutual information ↓
Host + latent ETKF (8 members)	295.4926	0.5709
Host + latent ETKF (100 members)	105.9172	0.2477
Host + latent ETKF (500 members)	19.6006	0.0622

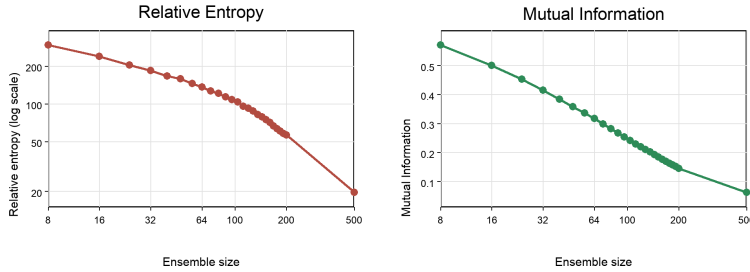


Figure 6: Moving MNIST latent-ETKF information metrics. Relative entropy measures the information discrepancy between forecast and analysis distributions, and mutual information measures how much uncertainty the observation removes from the latent state.

Table 1 and Figure 6 show that observation reduces the information gap. With 8 ensemble members, relative entropy is 295.4926 and residual mutual information is 0.5709. With 500 members, these values decrease to 19.6006 and 0.0622. In information space, the model moves from a self-contained forecast toward the observed ground-truth rollout.

B Additional Qualitative Results

Qualitative temporal samples: cases 3-4

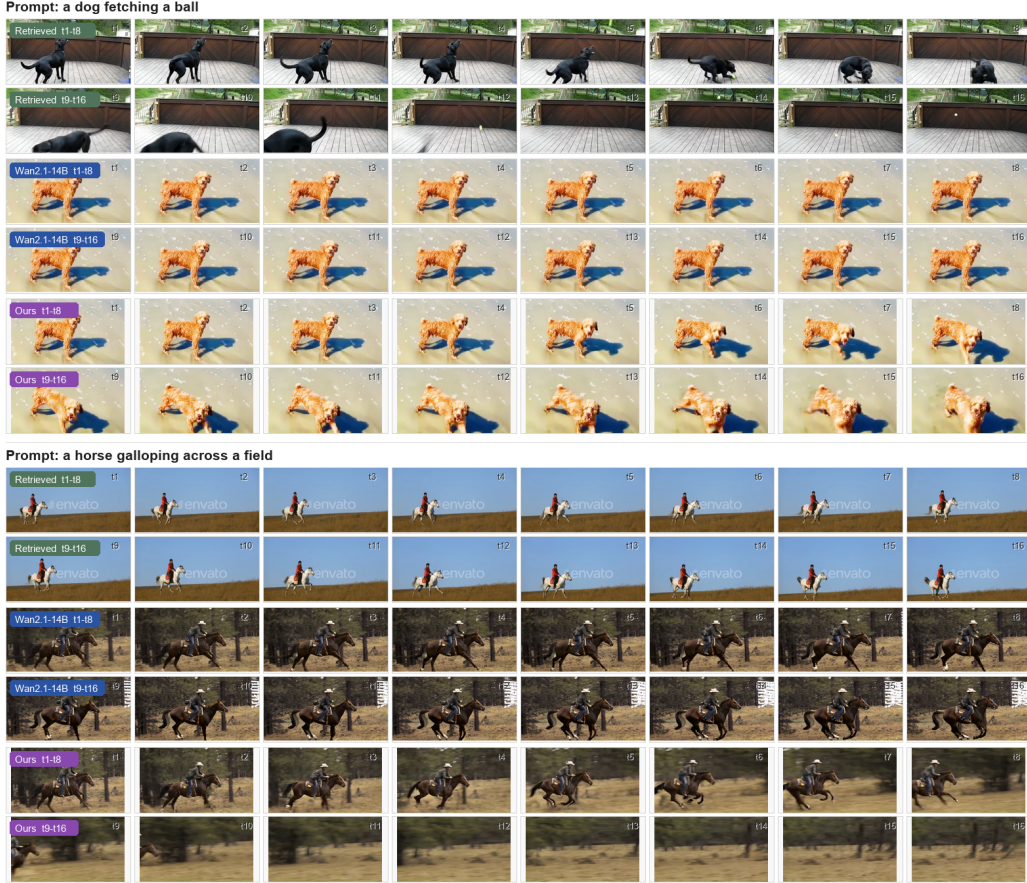


Figure 7: Temporal-sample qualitative comparison for cases 3–4. The retrieved rows show the real-video observation supplied to Video-as-Prompt, and the generated rows show Wan2.1-14B baseline inference and our retrieved-video inference at the same 16 normalized temporal positions.

Cases 3–4 show a similar pattern under animal motion. The dog sequence benefits from stronger body turning and approach motion, while the horse sequence becomes more energetic but loses framing in later samples. These failures indicate that motion correction alone is not sufficient for full spatial consistency.

Qualitative temporal samples: cases 5-6

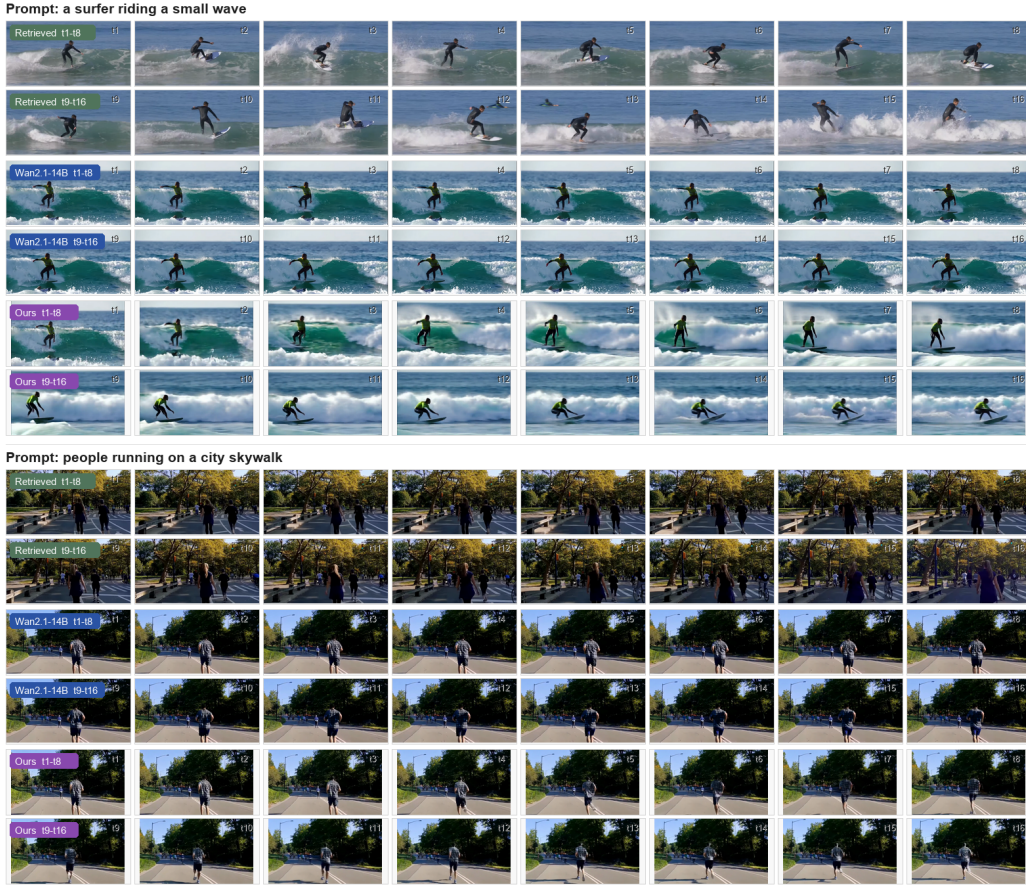


Figure 8: Temporal-sample qualitative comparison for cases 5–6. The two-row temporal layout makes scene-level motion visible while keeping each frame large enough to inspect.

Cases 5–6 are the most favorable examples for the observation-aware path. Surfing and running both require global scene motion, so the retrieved trajectory supplies useful temporal evidence that the image-only baseline tends to underuse. The outputs become more dynamic, although some blur and drift remain.

Table 2: Compact qualitative observations from the six-case pilot.

Case	Wan2.1-14B baseline	Ours + retrieved video observation
Basketball	Clean court and player; motion remains local.	Clearer shooting and follow-through, with some appearance drift.
Swing	Stable face, ropes, and background; modest forward motion.	Stronger swing depth and camera motion, but blur and weaker identity.
Dog	Mostly stationary dog; stable but weak dynamics.	More body turning and approach motion, with softer detail.
Horse	Plausible gallop pose; conservative motion.	More energetic lateral tracking, but late samples lose framing.
Surfing	Readable wave scene with moderate motion.	Stronger board, wave, and camera motion, with blur and subject drift.
Running	Foreground runner moves while road and crowd stay static.	Stronger forward and global scene motion, with mild blur.

C Related Work

Trinity consistency for world models. Wei et al. [23] argue that a general world model should be defined through modal, spatial, and temporal consistency. We adopt this trinity as the concep-

tual frame for the paper. Modal consistency connects prompts and visual observations to a shared semantic state. Spatial consistency preserves geometry, identity, and layout across frames and view-points. Temporal consistency governs causal evolution and motion. Our DA framing focuses on the intersection of spatial and temporal consistency: a retrieved real-video trajectory is treated as an observation that can stabilize the DiT forecast state.

Video generation as world modeling. Latent world models compress observations into a state representation and then evolve that representation forward with a learned dynamics model. This design underlies both classical latent control architectures and modern generative world models. Ha and Schmidhuber [7] made the paradigm explicit by separating representation, memory, and control, while recent surveys argue that modern video generators are increasingly being interpreted through a world-model lens [8, 22]. Under this lens, an image-to-video DiT is not only a renderer but also a latent transition model: it predicts how a scene state should evolve under a prompt, intent, or task description.

Spatial consistency and geometry-aware memory. A growing body of work attacks spatial inconsistency directly. World-consistent Video Diffusion uses explicit 3D coordinate frames to supervise 3D-consistent generation [27]. GEN3C conditions generation on a 3D cache to improve world-consistent video generation with precise camera control [15]. Video world models with long-term spatial memory store and retrieve geometry-grounded memory to prevent scene forgetting during revisits [25]. Other memory-based systems, including WorldPack and EvoWorld, improve spatial consistency through compressed or explicit memory mechanisms [13, 21]. These works motivate our use of one retrieved real-video trajectory as a spatial-temporal observation source.

State drift, memory, and physical plausibility. A related 2025–2026 line of work attacks temporal drift, weak state persistence, and physically implausible dynamics by strengthening memory, context, or closed-loop interaction. Examples include retrieval- or state-conditioned video world models [5, 24], embodied rollouts with mixture-of-experts or closed-loop training [17, 26], and persistent or embodiment-aware latent states [29, 6]. These works demonstrate that better state representations and memory mechanisms are critical, but they still largely handle external evidence through conditioning, retrieval augmentation, or latent persistence rather than through an explicit forecast–observation–analysis update inside the transition model.

Retrieval as external world memory. Retrieval and external memory have become common tools for improving video realism and long-horizon coherence. Retrieval-augmented generation appears in systems such as Peruzzo et al. [14] and Chen et al. [5], while explicit memory banks or persistent scene stores are central to Oshima et al. [13], Wang et al. [21], and Wu et al. [24]. In most of these methods, retrieved evidence is used as conditioning context, as a state prompt, or as an auxiliary memory. Our formulation is deliberately narrower and more explicit: retrieved evidence is treated as an observation of real-world dynamics. The retrieval memory is therefore an empirical world-memory bank, and the DiT must assimilate this observation into its latent forecast rather than merely append it as context.

Multimodal retrieval support. The formulation depends on an observation bank that is both prompt-annotated and large enough to supply semantically related real clips. OpenVid-1M is one practical choice for this role because it provides over one million text-video pairs and a curated high-definition subset [12]. For retrieval and prompt-alignment scoring we use LanguageBind, which places text, image, and video features into a shared representation space [30].

Video-as-prompt control. Video-as-Prompt (VAP) frames semantic-controlled video generation as in-context generation: a reference video provides the desired semantics, and a reference image provides the visual anchor [1]. We adapt this structure through retrieval-guided fine-tuning: the retrieved OpenVid clip becomes the reference video, and the input image remains the reference image. Our contribution is not a new attention mechanism; it is the DA interpretation, retrieval wrapper, and fine-tuning pipeline that treat the reference video as an observation trajectory from world memory and read the forecast-to-observation token interaction as an analysis update for the DiT transition state.

Data assimilation with learned and generative priors. DA provides a principled answer to sequential state estimation under model error and partial observations. The broader DA literature has already argued that latent and reduced-order spaces are natural places to fuse machine learning with estimation [3]. More recent work combines DA with learned priors or generative models, including sparse-observation weather DA [11], derivative-free ensemble guidance for diffusion models [28], stochastic-interpolant DA [4], and diffusion or flow-based perspectives on posterior correction and sequential warm starts [9, 16, 10]. These papers show that DA-style correction remains useful even when the predictive prior is highly expressive. However, they are not focused on video world models with retrieved visual observation memory. That gap is where our method sits.